

AI Phishing Detection Platform

Week 1 – Planning & Setup

1. Project Overview

The **AI Phishing Detection Platform** is a web-based cybersecurity application designed to help users identify potentially malicious or phishing-related URLs, emails, and messages. The platform analyzes suspicious content and provides risk levels, safety warnings, threat information, and detection history. The project focuses on practical full-stack development, cybersecurity concepts, API integration, and dashboard development.

2. Problem Statement

Many internet users find it difficult to identify fake or malicious websites, emails, and messages. Phishing attacks are designed to trick users into sharing sensitive information such as passwords, banking details, OTPs, and personal information.

The proposed system will help users verify suspicious links, analyze email content, detect possible phishing attempts, and improve cybersecurity awareness.

3. Project Objectives

The main objectives of the project are:

- To build a web-based phishing detection platform.
- To analyze suspicious URLs.
- To analyze email and message content.
- To generate risk analysis reports.
- To store scan history.
- To provide a user dashboard.
- To improve cybersecurity awareness.

4. Project Requirements

4.1 User Requirements

The system should allow users to:

- Register and log in.
- Scan suspicious URLs.
- Analyze email content.
- Analyze suspicious messages.
- View risk levels.
- View safety recommendations.
- View scan history.
- View detection reports.
- Access the dashboard.

4.2 URL Scanner Requirements

The URL scanner should analyze:

- HTTPS usage
- Suspicious keywords
- Domain structure
- Blacklisted URLs
- URL reputation using security APIs

The system should generate:

- Risk Level
- Threat Status
- Safety Recommendation

4.3 Email Analysis Requirements

The email analyzer should check for:

- Urgent or threatening language
- Suspicious links
- Fake security warnings
- Credential requests
- Scam-related keywords

4.4 Risk Analysis Requirements

The initial system will use:

- Rule-based detection
- Keyword analysis
- URL reputation
- Security APIs

The system will generate a risk percentage, suspicion level, and warning messages.

4.5 Dashboard Requirements

The dashboard should display:

- Total scans
- Safe URLs
- Suspicious URLs
- Scan history
- Detection statistics

4.6 Admin Requirements

The admin should be able to:

- Manage users
- View detection logs
- Monitor reports
- Manage suspicious keywords
- View analytics
- Track suspicious activity

5. Phishing Detection Research

URL Detection

The system can identify suspicious URLs using indicators such as:

- HTTPS usage
- Suspicious keywords
- Domain structure
- Blacklisted URLs

- URL reputation

Email Detection

The system can identify suspicious email content using:

- Urgent language
- Threatening messages
- Suspicious links
- Fake security warnings
- Credential requests
- Scam-related keywords

Detection Approach

Instead of training a complex machine-learning model from scratch, the project will initially use **rule-based detection, keyword analysis, security APIs, and URL reputation services**. This approach is intended to keep the project practical and beginner-friendly.

6. Technology Stack

Component	Technology
Frontend	Next.js, React.js, Tailwind CSS, Framer Motion
Backend	Node.js, Express.js
Database	Supabase
Authentication	JWT Authentication / Clerk
Security APIs	VirusTotal API, Google Safe Browsing API
Version Control	Git, GitHub

7. System Architecture

User



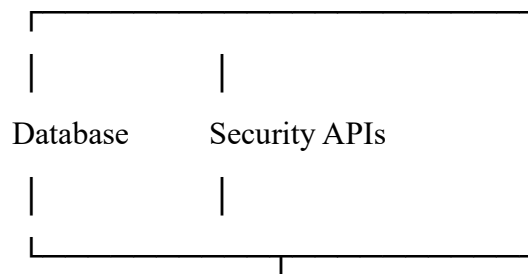
Frontend

(Next.js + React)



Backend

(Node.js + Express)



Database

Security APIs



Risk Analysis



Dashboard

8. Project Workflow

User submits URL / Email / Message



Frontend receives input



Backend processes request



Rule & Keyword Based Analysis



URL/API Security Check



Risk Score Generated



Result shown to the User



Scan saved in History